IE N5 CHAPTER 02 TUT 01

1. **QUESTION 2: POWER SUPPLIES**

2.1 Calculate the value of the second capacitor in the RC-π-filter circuit if the following values are known:

$V'_{R(RMS)}$ = 0,8 V $V_{R(RMS)}$ = 1,5 V R = 20 Ω
f = 50 Hz before full-wave rectification (6)

2.2 Draw a neat, labelled circuit diagram of an over-voltage protection circuit. (3)

2.3 Describe how the protection circuit in QUESTION 2.2 will protect the load of the circuit. (3)

2.4 A 600 mW 12 V Zener diode is used in a voltage reference source.

If the maximum supply voltage is 18 V, calculate the value of the series resistor in order to protect the Zener diode. (4)
[16]

2.

QUESTION 2: POWER SUPPLIES

2.1 An RC-π-filter is connected to a half-wave power supply which is provided with a 60 Hz supply. The serial resistor is 10 Ω, the second filter capacitor is 250 µf and the input ripple voltage is 5 V.

Calculate the ripple factor for a 15 V DC output. (5)

2.2 2.2.1 What is the main function of a voltage-multiplier circuit? (2)

2.2.2 Briefly give the disadvantage of a voltage-multiplier circuit. (2)

2.3 Draw a neat, labelled circuit diagram of a double-voltage power supply which can supply both positive and negative voltages to operational amplifiers. (8)
[17]

3.

QUESTION 2: POWER SUPPLIES

2.1 Calculate the value of the load resistor R_L if the following values of a capacitor filter are given:

V_m = 15 V; f = 50 Hz after half-wave rectification; C = 470 µf and I_{dc} = 10 mA (6)

2.2 Draw the basic circuit diagram that provides overcurrent protection in a power supply and briefly describe how the circuit operates. (6)

2.3 Calculate the drop in output voltage if a 15 V power supply (no-load voltage) has a voltage regulation of 5% as soon as it is connected to a load. (3)
[15]

4.

QUESTION 2: POWER SUPPLIES

2.1 Calculate the value of the load resistor R_L if the following values of a capacitor filter are known:

V_M = 20 V
f = 50 Hz after half-wave rectification
C = 470 µf
I_{DC} = 15 mA. (6)

5.

QUESTION 2: POWER SUPPLIES

2.1 The output voltage across the first capacitor of a LC-π-filter half-wave rectifier is 150 V DC with a 10 V ripple at a frequency of 110 Hz.

Calculate the output ripple factor of the filter if L = 2 H, R = 250 Ω and C_2 = 6 µf. The filter is connected to a 5 kΩ load. (7)

2.2 Draw a neat, labelled block diagram of a serial voltage regulator and describe the function of each block. (10)
[17]

6.

QUESTION 2: POWER SUPPLIES

2.1 Calculate the percentage voltage regulation of a power source if the ripple factor and ripple voltage without a load are 3% and 0,967 V respectively and the full load voltage is 11,8 V. (4)

2.2 Briefly explain how the total voltage drop across a half-wave voltage trippler is obtained. (4)

2.3 Draw a neat, labelled circuit diagram of a complete power supply that includes the following:

- Transformer
- Bridge rectifier
- RC-π-filter
- Series voltage regulator with an operational amplifier and feedback network
- Overcurrent protection circuit (8)

[16]

7. **QUESTION 2: POWER SUPPLIES**

2.1 Briefly describe how the total drop across a full-wave doubler is obtained. (4)

2.2 The following values of a LC-π-filter are known: I'_{DC} = 2 A; R_1 = 5 Ω; R_L = 10 Ω; L = 0,1 H; X_{C2} 20 Ω; $V_{R(RMS)}$ = 1,5 V and 50 Hz full-wave rectification.

Calculate the following:

2.2.1 Output voltage across the load (2)

2.2.2 Output ripple voltage (5)

8. **QUESTION 2: POWER SUPPLY**

2.1 Calculate the drop in the output voltage if a 20,5 V power supply (NO-load voltage) has a voltage regulation of 6% as soon as it is connected to a load. (3)

2.2 Briefly describe how the total voltage drop across a half-wave tripler is obtained. (4)

2.3 Draw a neat, labelled circuit diagram of a serial voltage regulator which includes a closed-loop operation. (5)

2.4 Draw a neat, labelled circuit diagram of a 12 V power supply that uses an integrated positive voltage regulator. (4)

[16]

9. **QUESTION 2: POWER SUPPLIES**

2.1 The following values of a RC-filter are known: X_{C2} = 26,526 Ω; R_L = 1,2 kΩ; R = 450 Ω; r' = 1,3% and $V'_{R(RMS)}$ = 0,118 V at C_2 f = 50 Hz. The supply is rectified by a half-wave rectifier.

Calculate:

2.1.1 V_{DC} across the first capacitor (4)

2.1.2 $V_{R(RMS)+}$ across the first capacitor (3)

2.1.3 r across the first capacitor (2)

2.2 Briefly define the term *voltage regulation*. (2)

2.3 Draw a neat, labelled block diagram of a double power supply, which can supply both positive and negative voltages to operational amplifiers. (6)

[17]